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Part A

1. Define the following terms in measurement.

i) Accuracy ii) Resolution iii) Precision [3 marks] (December 2021)

(i) Accuracy: It is the degree of closeness with which the reading approaches the true value of the quantity to be measured.

(ii) Resolution: If the input is slowly increased from some arbitrary input value(nonzero), it will again be found that output does not change at all until a certain increment is exceeded. This increment is called resolution. Smallest measurable i/p change.

(iii) Precision: It is the measure of repeatability or reproducibility i.e., given a fixed value of a quantity, precision is a measure of the degree of agreement within a group of measurements.

2. What are the different standards of measurement? [3 marks] (December 2021)

A standard of measurement is a physical representation of a unit of measurement.

The classifications of standards are

1. International standards 2. Primary standards 3. Secondary standards 4. Working standards 5. Current standards 6. Voltage standards 7. Resistance standards

The international standards are defined by international agreements. They represent certain units of measurement to the closest possible accuracy that production and measurement technology allows.

The highest standard of either a base unit or a derived unit is called a primary standard. The primary standards are maintained by national standards laboratories in different places of the world.

Secondary standards are the basic reference calibrated standards designed and calibrated from the primary standards used in industrial measurement laboratories.

Working standards are the principal tools of a measurement laboratory

Explain the significance of measurements. [3 marks] (December 2020)

Measurement provides a standard for everyday things and processes. Measurement helps us to compare unknown quantities with known quantities. Measurement of a given quantity is essentially an act or the result of a comparison between the quantity (whose magnitude is unknown) and a predefined value. Measurement helps us make quantitative statements about how big, how long, how fast things are. Without measurement, the final product will be full of errors. From weight, temperature, length even time is a measurement and it does play a very important role in our lives.

ex 1: we use measurement to weigh our weights. A kilogram is a unit for weight measurement.

ex 2: we use measurement to know our height.

How drift affects the input-output relationship? [3 marks] (December 2020)

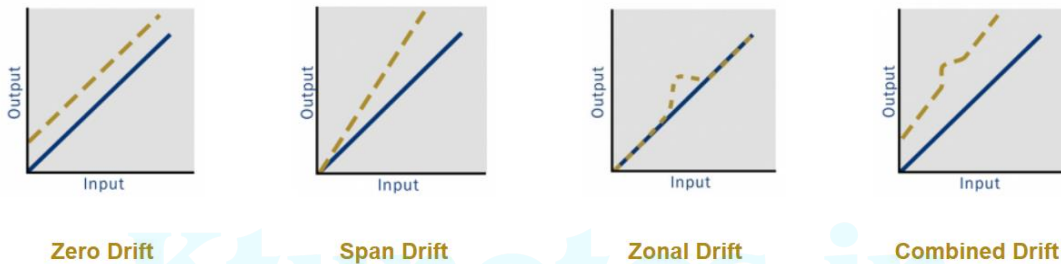
EET 203 Measurements & Instrumentation
Module 1: University Questions and Answers

Prepared by Dr. Renuka T K , EEE, MESCE

Drift is a measurement error caused by the gradual shift in a gauge's measured values over time. Although incorrect handling can accelerate it, nearly all measuring instruments will experience drift during their lifetime. If left unchecked, this shift can cause extensive measuring errors, safety hazards, and quality issues.

Short-term drift is a temporary effect on measured values caused by flexible factors such as thermal expansion, environmental interference, and vibrations. Once removed from its environment or allow to rest, the gauge's measured values will revert towards the calibrated values.

Long-term drift is typically caused by regular wear and tear or other naturally occurring damage. Long-term drift typically requires an adjustment to return measurements to the correct values. Since long-term drift typically develops at a consistent rate, it can be predicted and corrected before instruments move outside of tolerance.



Even though drift is unavoidable, proper care and maintenance is the best method of minimizing its development.

Explain with neat diagram the construction and principle of operation of a PMMC instrument. Derive the expression for deflection. [7 marks] (December 2021)

Construction of PMMC Instruments • Permanent Magnet Moving Coil or PMMC Instruments consists of following components

• Deflecting system • Magnet System • Control Spring • Damping • Pointer and Scale

Deflecting system:

- The moving coil made up of copper is wound with many turns on the rectangular Aluminum former. This Aluminum former is pivoted on the jewelled bearing. The coil can move freely in the magnetic field produced by the Permanent Magnet System. The current or voltage to be measured is passed through the coil. The coil experience a force and provide necessary deflection.

Magnet System:

- Simple U shaped permanent magnet made of Alcomax or Alnico is widely use in PMMC instruments. Theses magnets have high coercive force and can produce field of the order of 0.1 to 1 Wb/m².

Control Spring:

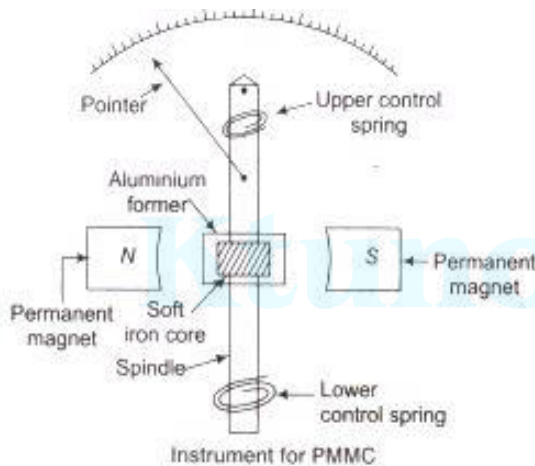
- The controlling torque in PMMC Instruments is provided by two control spring mounted on the jewel bearing.

Damping:

- Damping torque in PMMC instruments are produced by the movement of Aluminum former in the magnetic field of Permanent Magnet. Due to movement of Aluminum former an emf is induced resulting in eddy current which opposes the motion thereby provides a damping torque.

Pointer and Scale:

- The pointer is carried by the spindle and moves over a graduated scale.



Principle of Operation:

The principle on which a Permanent Magnet Moving Coil (PMMC) instrument operates is that a torque is exerted on a current-carrying coil placed in the field of a permanent magnet.

When DC supply is given to moving coil DC current flows through it.

When the current carrying coil is kept in magnetic field it experiences a force. This force produces torque and former rotates.

Pointer is attached with spindle & when former rotates pointer moves over the calibrated scale.

If ac is supplied reversing torque is produced this cannot produce continuous deflection therefore cannot be used in AC.

Torque Developed

- The general expression for force produced in the coil $F = NBI l \sin\theta$, Where N–Number of turns of coil, B–flux density in the air gap , l, b – the vertical and horizontal length of the side, I – current through the coil.

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Here Field is radial so $\theta=90^\circ$, Then $F = NBI l$

Torque T_d is Force * Distance

$$\text{i.e } T_d = NBI l * b = NBIA$$

Hence $T_d \propto I$, i.e $T_d = GI$, where G is a constant

Control Torque: The control on the movement of the pointer over the scale is provided by two spirally wound, phosphor-bronze springs S_1 and S_2 , one at each end of the spindle S .

- Sometimes these springs also conduct the current into and out of the coil.
- The control torque of the springs is proportional to the angle θ turned through by the coil. i.e $T_c = K\theta$

Where K = Spring constant, θ = Angular movement of coil.

- At steady state condition, deflecting and controlling torque shall be equal, i.e $T_d = T_c$

$$\Rightarrow GI = K\theta$$

$\Rightarrow \theta = (G/K)I$, Deflection directly proportional to I passing through the meter scale is uniform (linear)

Damping Torque: When the aluminium former (F) moves with the coil in the field of the permanent magnet, a voltage is induced, causing eddy current to flow in it. This current exerts a force on the former. By Lenz's law, this force opposes the motion producing it. Thus, a damping torque is obtained. Such damping is called eddy-current damping.

Derive the expression for deflection for spring-controlled attraction type moving iron instrument. Also, explain the type of damping provided in moving iron instruments.

[7 marks] (December 2021)

- Attraction type moving iron instrument consists of a flat disc of iron which is mounted on the spindle which carries the pointer. Damping is provided by air friction damping and control torque is provided with the help of spring.
- When current passes through the coil the magnetic field is developed which is stronger towards the inner side of the coil. The moving iron piece has a tendency to move from a position of a weaker field to a position of a stronger field. Hence it experiences a force of attraction & moves into the coil resulting in the deflection of pointer.

Torque Equation of Moving-Iron Instruments

- To deduce the expression for torque of a moving iron instrument, energy relation can be considered for a small increment in current supplied to the instrument. This results in a small deflection $d\theta$ and some mechanical work will be done. Let T_d be the deflecting torque.
- Therefore mechanical work done = torque $\times$ angular displacement

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- Due to the change in inductance there will be a change in the energy stored in the magnetic field.

$$= T_d \cdot d\theta$$

Let I be the initial current, L be the instrument inductance and θ is the deflection. If the current increases by dI then it causes the change in deflection $d\theta$ and the inductance by dL . In order to involve the increment dI in the current, the applied voltage must be increase by:

$$e = \frac{d\phi}{dt} = \frac{d}{dt}(LI) = I \frac{dL}{dt} + L \frac{dI}{dt}$$

The electrical energy supplied is $eIdt = I^2dL + ILdI$

The current is changes from I to $(I + dI)$, and the inductor L to $(L + dL)$

Therefore the stored energy changes from $= \frac{1}{2}I^2L$ to $\frac{1}{2}(I + dI)^2 (L + dL)$

Hence the change in stored energy $= \frac{1}{2}(I + dI)^2 (L + dL) - \frac{1}{2}I^2L$

As dI and dL are very small, neglecting the second and higher order terms in small quantities, this

becomes $ILdL + \frac{1}{2}I^2dL$

From the principle of conservation of energy,

Electrical energy supplied = Increase in stored energy + Mechanical work done

$$I^2dL + ILdI = ILdI + \frac{1}{2}I^2dL + T_d d\theta$$

$$\therefore T_d d\theta = \frac{1}{2}I^2dL$$

$$\text{or deflecting torque } T_d = \frac{1}{2}I^2 \frac{dL}{d\theta}$$

where T_d is in newton-metre, I is in ampere, L is in henry and θ is in radians.

The moving system is provided with control springs and in turn the deflecting torque T_d is balanced by the controlling torque $T_C = k \theta$

where k is the control spring constant (N-m/rad) and θ is the deflection in radians.

At final steady position, $T_C = T_d$

$$\text{or } k\theta = \frac{1}{2}I^2 \frac{dL}{d\theta}$$

$$\therefore \text{ deflection } \theta = \frac{1}{2} \frac{I^2}{k} \frac{dL}{d\theta} \quad (2.33)$$

Hence, the deflection is proportional to square of the rms value of the operating current. The deflection torque is, therefore, unidirectional whatever may be the polarity of the current.

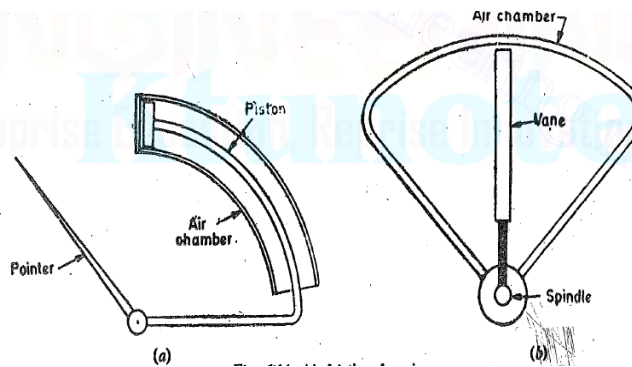
Air Friction Damping

In this method, a light aluminium piston is attached to the moving system and moves in an air chamber closed at one end.

The cross-section of this chamber may be either circular or rectangular.

The clearance between the piston and the sides of the chamber should be small and uniform.

If the piston is moving rapidly into the chamber, the air in the closed space is compressed and the pressure opposes the motion of the piston (and, therefore, of the whole moving system). If the piston is moving out of the chamber rapidly, the pressure in the closed space falls, and the pressure on the open side of the piston is greater than that on the opposite side. Motion is thus again opposed. Sometimes instead of a piston, a vane, mounted on the spindle of the moving system, moves in a closed-sector-shaped chamber.



What are the different torques needed for the proper operation of an indicating instrument?

[7 marks] (December 2021)

- Indicating instruments are used to indicate the quantity with respect to a pointer that is being measured at the time when it moves on a scale of an instrument. Some instruments consist of a pointer which moves over a calibrated scale. The moving system of the indicating instruments consists of three torques.

Deflecting Torque

- The deflecting torque is produced by utilizing any of the effects like magnetic, electrostatic, electrodynamics, thermal or inductive etc. Thus the deflecting system of an instrument converts the electric current or potential into mechanical force called deflecting force. The deflecting system thus acts as prime mover responsible for deflection of pointer.

This deflecting torque causes the moving system to move from its zero position or when the instrument is disconnected from the supply.

Controlling Torque

This force is required in order that the current produces deflection of pointer proportional to its magnitude. This torque faces the deflecting torque and increases with the deflection of the moving system. The pointer moves on its rest position when these two opposing torques are equal. Another function of controlling system is to bring moving system back to zero when the force causing the moving system to deflect is removed. During the absence of a restoring torque, the pointer doesn't return to its zero position when the supply is removed from the instrument.

The controlling or balancing torque in indicating instruments is obtained by two methods - spring control and gravity control.

Damping Torque

Damping Systems. The damping torque should be of such a magnitude that the pointer quickly comes to its final steady position, without overshooting.

If the instrument is underdamped, the moving system will oscillate about the final steady position with a decreasing amplitude and will take some time before it comes to rest.

When the moving system moves rapidly but smoothly to its final steady position, the instrument is said to be critically damped or dead beat.

If the damping torque is more than what is required for critical damping, the instrument is said to be overdamped. In an overdamped instrument, the moving system moves slowly to its final steady position in a lethargic fashion. The readings are very tedious to take.

Damping is generally provided by air friction, fluid friction, or eddy current.

The coil of a measuring instrument has a resistance of 1 Ohm and the instrument has a full scale deflection of 250V when a resistance of 4999 Ohm is connected in series with it. Find

(i). The current range of the instrument when used as an ammeter with the coil connected across a shunt of 1/499 Ohm.

(ii). The shunt resistance for the instrument to give a full-scale deflection of 50 A.

[6 marks] (December 2021)

$$I_m = 250 / (1 + 4999) = 0.05 \text{ A}$$

$$R_{sh} = R_m / (N - 1) \quad \text{where } N = I / I_m, \quad R_m = 1 \text{ ohm}$$

$$1/499 = 1 / (N - 1) \quad \text{i.e } N - 1 = 499$$

$$I / 0.05 = 499$$

(i) So The current range of the instrument when used as an ammeter when a shunt is connected

$$I = 499 \times 0.05 = 24.95 \text{ A} \approx 25 \text{ A}$$

$$(ii) R_{sh} = R_m / (N-1) = 1 / ([50/0.05] - 1) = 1/999 \text{ Ohm} = 0.001 \text{ Ohm}$$

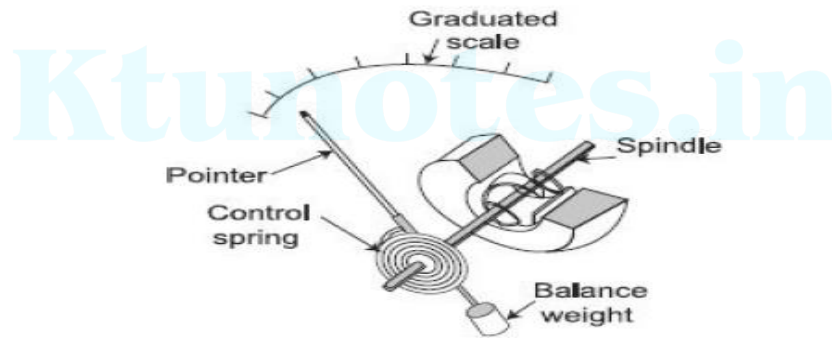
What is controlling force? Explain the various controlling systems used in an indicating instrument.

[6 marks] (December 2020)

Controlling force is required in order that the current produces deflection of pointer proportional to its magnitude. This torque faces the deflecting torque and increases with the deflection of the moving system. The pointer moves on its rest position when these two opposing torques are equal. Another function of the controlling system is to bring the moving system back to zero when the force causing the moving system to deflect is removed.

During the absence of a restoring torque, the pointer doesn't return to its zero position when the supply is removed from the instrument. The controlling or balancing torque in indicating instruments is obtained by two methods.

Spring control:



A spiral hairspring made of some non-magnetic material like phosphor bronze is attached to the moving system of the instrument as shown in the figure

With the deflection of the pointer, the spring is twisted in the opposite direction. This twist in the spring provides the controlling torque. Since the torsion torque of a spiral spring is proportional to the angle of twist, the controlling torque (T_c) is directly proportional to the angle of deflection of pointer (θ) i.e. $T_c \propto \theta$. The pointer will come to rest at a position where controlling torque is equal to the deflecting torque i.e. $T_c = T_d$

To give a controlling torque which is directly proportional to the angle of deflection of the moving system, the number of turns on the spring should be fairly large, so that the deflection per unit length is small.

The stress in the spring must be limited to such a value that there is no permanent set.

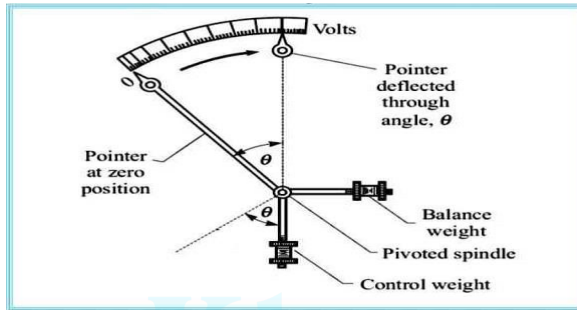
Suppose that a spiral spring is made up of a total length L m of strip whose cross section is rectangular, the radial thickness being t m and the depth b m.

Let E be Young's modulus (N/m^2) for the material of the spring. Then, if θ radians be the deflection of the moving system to which one end of the spring is being attached, the expression for the controlling torque is

$$T_c = (Ebt^3/ 12L) \theta$$

Thus, controlling torque $\propto \theta \propto$ instrument deflection.

Gravity Control



In this method, a small weight is attached to the moving system, which provides necessary controlling torque.

In the zero position of the pointer, the control weight hangs vertically downward and therefore provides no controlling torque.

However, under the action of deflecting torque, the pointer moves from zero position and control weight moves in opposite direction.

Due to gravity, the control weight would tend to come in original position (i.e. vertical) and thus provides an opposing or controlling torque.

The pointer comes to rest at a position where controlling torque is equal to the deflecting torque. In this method, controlling torque (T_c) is proportional to the sin of angle of deflection (θ) i.e. $T_c \propto \sin \theta$.

Because in this method controlling torque (T_c) is not directly proportional to the angle of deflection (θ) but it is proportional to $\sin \theta$ therefore, gravity control instruments have non-uniform scales; being crowded in beginning.

Explain the various methods for producing damping torque with neat figures.

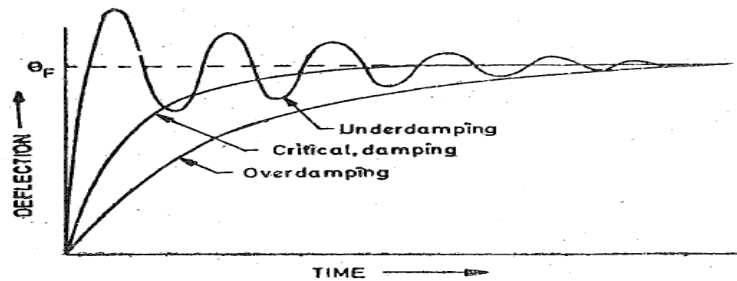
[8 marks] (December 2020)

Damping Systems. The damping torque should be of such a magnitude that the pointer quickly comes to its final steady position, without overshooting.

If the instrument is underdamped, the moving system will oscillate about the final steady position with a decreasing amplitude and will take some time before it comes to rest.

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Air Friction Damping

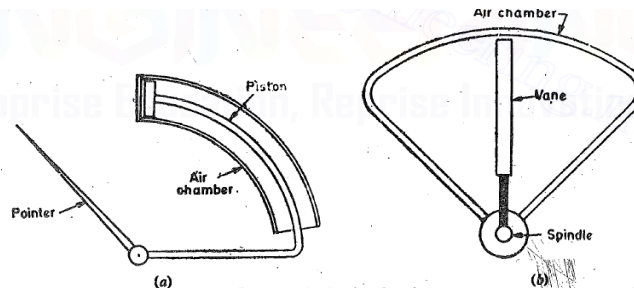
In this method, a light aluminium piston is attached to the moving system and moves in an air chamber closed at one end.

The cross-section of this chamber may be either circular or rectangular.

The clearance between the piston and the sides of the chamber should be small and uniform.

If the piston is moving rapidly into the chamber, the air in the closed space is compressed and the pressure opposes the motion of the piston (and, therefore, of the whole moving system). If the piston is moving out of the chamber rapidly, the pressure in the closed space falls, and the pressure on the open side of the piston is greater than that on the opposite side. Motion is thus again opposed.

Sometimes instead of a piston, a vane, mounted on the spindle of the moving system, moves in a closed-sector-shaped chamber.

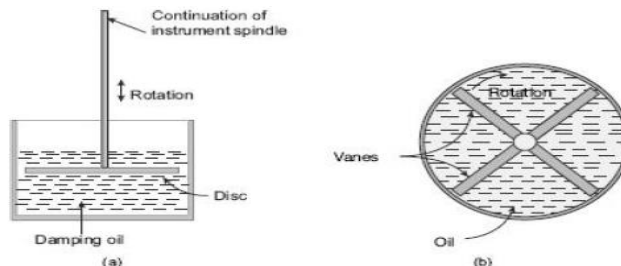


Fluid Friction Damping

In this type of damping, a light vane, attached to the spindle of the moving system, dips into a pot of damping oil and should be completely submerged by the oil.

The frictional drag in the disc is always in the direction opposing motion. There is no friction force when the disc is stationary.

In the second system increased damping is obtained by the use of vanes.



Eddy Current Damping

- When a sheet of conducting material moves in a magnetic field so as to cut through lines of force, eddy currents are set up in it and a force exists between these currents and the magnetic field, which is always in the direction opposing the motion.
- The force is proportional to the magnitude of the current and to the strength of the field.
- The magnitude of the current is proportional to the velocity of movement of the conductor, and thus, if the magnetic field is constant, the damping force is proportional to the velocity of the moving system and is zero when there is no movement of the system.

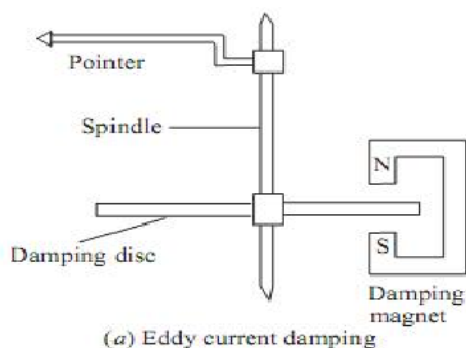


Figure shows a metallic disc rotating in the field of a permanent magnet. In this type of damping, a copper or aluminium disc is attached to the moving system of the pointer and permanent magnets are attached to the poles. When the pointer moves the copper or aluminium disc cuts the magnetic flux between the poles of a permanent magnet and eddy current is produced with the flow and produces the damping force in a required direction.

With neat sketches explain the construction of a PMMC instrument.

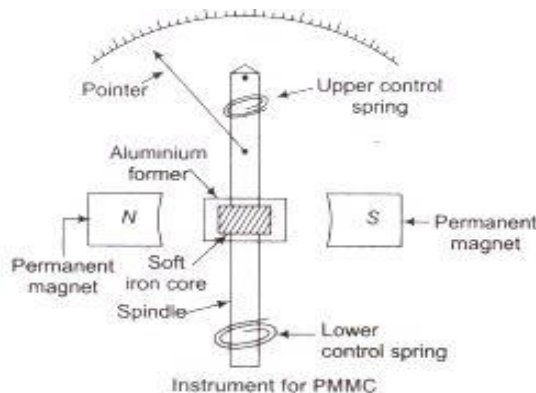
[8 marks] (December 2020)

Permanent Magnet Moving Coil (PMMC) Instrument –(DC)

Permanent Magnet Moving Coil or PMMC Instruments are the most accurate type for the measurement of DC current or voltage.

Construction of PMMC Instruments

Permanent Magnet Moving Coil or PMMC Instruments consists of following components



- Deflecting system
- Magnet System
- Control Spring
- Damping
- Pointer and Scale

The principle on which a Permanent Magnet Moving Coil (PMMC) instrument operates is that a torque is exerted on a current-carrying coil placed in the field of a permanent magnet

The coil C has a number of turns of thin insulated wires wound on a rectangular aluminium former F.

The frame is carried on a spindle S mounted in jewel bearings J1, J2.

A pointer PR is attached to the spindle so that it moves over a calibrated scale.

The whole of the moving system is made as light weight as possible to keep the friction at the bearing to a minimum.

The coil is free to rotate in air gaps formed between the shaped soft-iron pole piece (pp) of a permanent magnet PM and a fixed soft-iron cylindrical core IC .

The core serves two purposes; (a) it intensifies the magnetic field by reducing the length of the air gap, and (b) it makes the field radial and uniform in the air gap. Thus, the coil always moves at right angles to the magnetic field .

Modern permanent magnets are made of steel alloys which are difficult to machine. Soft-iron pole pieces (pp) are attached to the permanent magnet PM for easy machining in order to adjust the length of the air gap.

A soft-iron yoke (Y) is used to complete the flux path and to provide shielding from stray external fields.

Deflecting system:The moving coil made up of copper is wound with many turns on the rectangular Aluminum former. This Aluminum former is pivoted on the jewelled bearing. The coil can move

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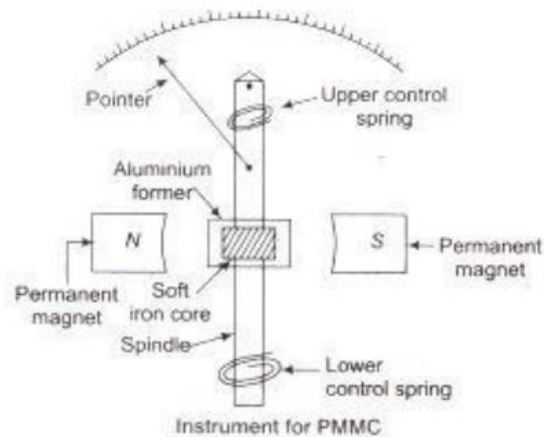
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Pointer and Scale: The pointer is carried by the spindle and moves over a graduated scale.

“PMMC instruments have uniform linear scale”. Justify.

[6 marks] (December 2020)

The principle on which a Permanent Magnet Moving Coil (PMMC) instrument operates is that a torque is exerted on a current-carrying coil placed in the field of a permanent magnet. When DC supply is given to moving coil DC current flows through it. When the current carrying coil is kept in magnetic field it experiences a force. This force produces torque and former rotates. Pointer is attached with spindle & when former rotates pointer moves over the calibrated scale



The general expression for force produced in the coil $F = NBI \sin\theta$, Where N–Number of

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turns of coil, B – flux density in the air gap , l , b – the vertical and horizontal length of the side, I – current through the coil.

Here Field is radial so $\theta=90^\circ$, Then $F = NBI$

Torque T_d is Force * Distance

i.e $T_d = NBI l * b = NBI A$

Hence $T_d \propto I$, ie $T_d = GI$, where G is a constant

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